RoO Input–Output Codes for Five PTAs: Results

19 October 2026 · August Liu. Companion to the coding workflow document (general coding rules) and the variable explanation file.

1. Data

- Five agreements: ASEAN–Japan, Brunei–Japan, Australia–US, CAFTA and Bahrain–US, all in HS2002. Each row is one (restricted input, output) pair at HS6 from the agreement's product-specific rules: 1,099,332 rows in all.
- One Stata file per agreement (*<PTA>_coded_2026_10_19_v2.dta*), with the same 29 variables.
- The four new agreements contain 358,048 unique pairs; 98,336 of them are also in ASEAN–Japan. Every pair is coded by the rules of the workflow document, so a pair has the same code in every agreement.

2. Main results

Table 1. Input–output links by agreement

Agreement	Rows	DV = 1	Share	DV_strict = 1	DV_excl_remelt = 1	Uncertain	Uncertain / DV = 1
ASEAN–Japan	115,764	53,949	46.6%	49,811	53,526	4,138	7.7%
Brunei–Japan	83,491	36,117	43.3%	32,929	35,939	3,188	8.8%
Australia–US	333,324	222,942	66.9%	195,187	222,121	27,755	12.4%
CAFTA	321,967	215,947	67.1%	193,088	215,133	22,859	10.6%
Bahrain–US	244,786	185,904	75.9%	165,928	185,448	19,976	10.7%

DV = 1: the input can be used to make the output (main variable). DV_strict counts only certain links (DV × (1 − uncertain)). DV_excl_remelt sets direct remelting (route = recycle_direct, for example steel rod → steel ingots) to 0.

3. Uncertain pairs

Table 2. Pairs kept at DV = 1 with uncertain = 1, by reason

Reason	ASEAN–Japan	Brunei–Japan	Australia–US	CAFTA	Bahrain–US
Minor constituent of a garment or made-up: other fibre (not cotton or man-made) or other construction	0	273	16,863	14,512	16,083
Minor constituent of a yarn or fabric: other fibre (not cotton or man-made)	118	190	2,243	1,250	2,243
The two AI coders (or the model and the textile rules) disagree	0	980	4,983	4,086	135
The two AI coders disagree: intermediate sibling or backward	0	27	1,079	558	92
Open case in the ASEAN–Japan labels	4,016	1,714	2,583	2,449	1,423
Possible feed (crustaceans → rock lobster)	4	4	4	4	0
Total uncertain	**4,138**	**3,188**	**27,755**	**22,859**	**19,976**

Most uncertain pairs in the three US agreements are minor constituents: a yarn or fabric of a less common fibre (wool, silk, flax, ramie, coarse animal hair) in a garment, made-up, yarn or fabric of another fibre, or a fabric of the same fibre but another construction (woven fabric in a knitted garment). They are frequent there because the US yarn-forward rules list almost every yarn and fabric as a restricted input of each garment. Textile pairs (chapters 50–63 on both sides) make up ASEAN–Japan 44.1%, Brunei–Japan 43.4%, Australia–US 74.6%, CAFTA 75.2%, Bahrain–US 99.2% of the rows. Cotton and man-made fibres as minor constituents are coded as certain links.

4. How the codes were obtained

Table 3. Rows by coding method

Method	ASEAN–Japan	Brunei–Japan	Australia–US	CAFTA	Bahrain–US	Unique pairs, four new PTAs
ASEAN–Japan labels (hand-coded, revised)	115,764	0	0	0	0	–
Copied from ASEAN–Japan (step 7a)	0	64,595	91,602	88,678	50,871	98,336
Textile rules (step 7e)	0	7,854	185,894	179,583	180,589	187,479
Two AI coders (steps 2–6)	0	6,285	41,172	39,364	1,853	57,404
Model, confirmed by the textile rules (step 7c)	0	2,221	9,272	9,128	9,272	9,340
Self-pairs (step 2)	0	2,101	3,311	3,206	128	3,388
Model and textile rules disagree (step 7d)	0	435	2,073	2,008	2,073	2,101
Total	**115,764**	**83,491**	**333,324**	**321,967**	**244,786**	**358,048**

ASEAN–Japan was coded by hand and revised under the same conventions; it serves as the master list for exact copies. The model bands were used only for textile pairs confirmed by the textile rules (section 5). Where the model and the textile rules disagreed, the rule was applied when it found a route; the remaining pairs were settled with the audited sample.

5. Reliability

- **Consistency across agreements.** 327,518 pairs appear in two or more agreements; DV, uncertain, route and uncertain reason are identical in every agreement for all of them (0 differences).
- **Two AI coders.** Claude and Codex coded 56,616 pairs independently (before the scope conventions of 19 October). They agreed on 70.9% (Cohen's kappa 0.39). Agreement was high for plant products and fats (kappa 0.87) and textiles (0.86), and low for chemicals (0.14), plastics (0.22) and food preparations (0.27): mostly the scope questions (intermediate siblings, chemicals → wastes, minerals → cement, possible uses) that the conventions now settle. Remaining disagreements are kept at DV = 1 with uncertain = 1.
- **Model.** With whole chapter pairs held out, the model reaches an AUC of 0.83. On 8,516 non-textile pairs in its confidence bands, re-coded by both AI coders, the ≥ 0.9 band was right 89.6% of the time and the ≤ 0.1 band 80.8%. The bands were therefore used only for textile pairs confirmed by the rules; all other band pairs went to the AI coders.
- **Textile rules.** On 50,249 textile pairs that are also in ASEAN–Japan, the rules agree with the ASEAN–Japan labels for 92.3% of pairs.
- **Independent AI audit.** A stratified sample of 295 pairs (by source, step and route) was coded by two new AI sessions that saw only the codebook and the pairs, not any earlier answer. August reviewed every pair where both auditors disagreed with the final code. The two auditors agree with each other on 78.0% of pairs (kappa 0.53). Their disagreements with the final codes showed groups of errors (intermediate siblings coded 0, sewing thread ↔ yarn, clothing trims → garments, minor yarns in fabrics, fish of different species), which were corrected by rule in all five agreements.

Table 4. Independent AI audit (295 pairs, weighted by stratum size)

Share of pairs where the final code matches …	Before corrections	After corrections
auditor 1 (new Claude session)	88.7%	97.3%
auditor 2 (new Codex session)	82.0%	90.6%
both auditors, where they agree (230 pairs)	89.6%	99.2%
pairs where both auditors disagree with the final code	25	5

The after-correction figures are measured on the same sample that revealed the errors, so they overstate accuracy. Of the 5 remaining differences, four are single pairs coded 0 by both original AI coders (glass smallwares → glass fibres; pickling preparations → fatty acids; paper-pulp filter blocks → coated paper; coated paper → cigarette paper) and one is already uncertain (pet food → oil-cake). This is an AI audit reviewed by a research assistant, not an expert hand-coding.

6. Revisions of 18–19 October 2026

Table 5. DV = 1 and uncertain pairs by version

Agreement	DV = 1, 18 Oct	DV = 1, 19 Oct (sent)	DV = 1, final	Uncertain, 18 Oct	Uncertain, 19 Oct (sent)	Uncertain, final
ASEAN–Japan	55,538	57,109	53,949	4,016	4,260	4,138
Brunei–Japan	38,152	39,114	36,117	6,182	6,480	3,188
Australia–US	204,601	225,466	222,942	95,030	108,885	27,755
CAFTA	198,721	218,471	215,947	95,626	103,335	22,859
Bahrain–US	165,095	185,509	185,904	80,358	98,174	19,976

- **18 October:** first complete coding of the four new agreements.
- **19 October (sent):** scope conventions (intermediate siblings, one synthetic fibre, chemicals → wastes, minerals → cement, possible uses, minor constituents, direct remelting) and the first audit corrections, applied to all five agreements, including the ASEAN–Japan labels. Pairs only moved from 0 to 1.
- **Final:** (i) fish and other seafood of different species are 0 unless the input could feed the output (3,160 ASEAN–Japan pairs moved from 1 to 0, and their copies in the other agreements); (ii) “of other textile materials” is read as in the HS (every fibre not named in its group), which added a few hundred links in each US agreement; (iii) cotton and man-made fibres as minor constituents are certain links, which cut the uncertain share of DV = 1 in the US agreements from about half to 11–12%.

7. Notes

- ASEAN–Japan pairs left open in the original hand coding (route = unresolved) are kept at DV = 1 with uncertain = 1, and the same pairs carry the same code in the other agreements.
- concord_flag = 1 marks 13 rows with a code that is not a valid HS2002 subheading (290400 in ASEAN–Japan, 640410 in Australia–US).
- The AI coders are Claude (Anthropic) and Codex (OpenAI); their codes and one-sentence reasons are kept in dv_ai1, dv_ai2, why_ai1 and why_ai2 where they coded the pair.